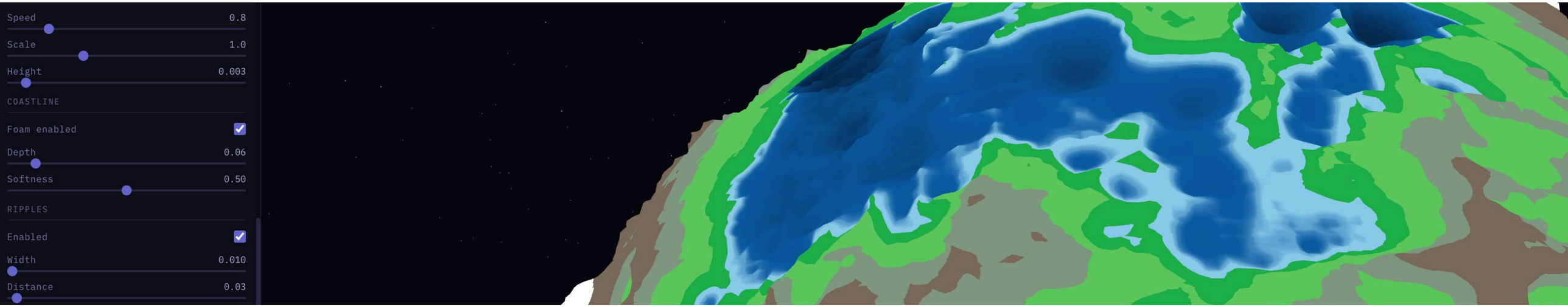
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Class 02

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HTML CSS Javascript

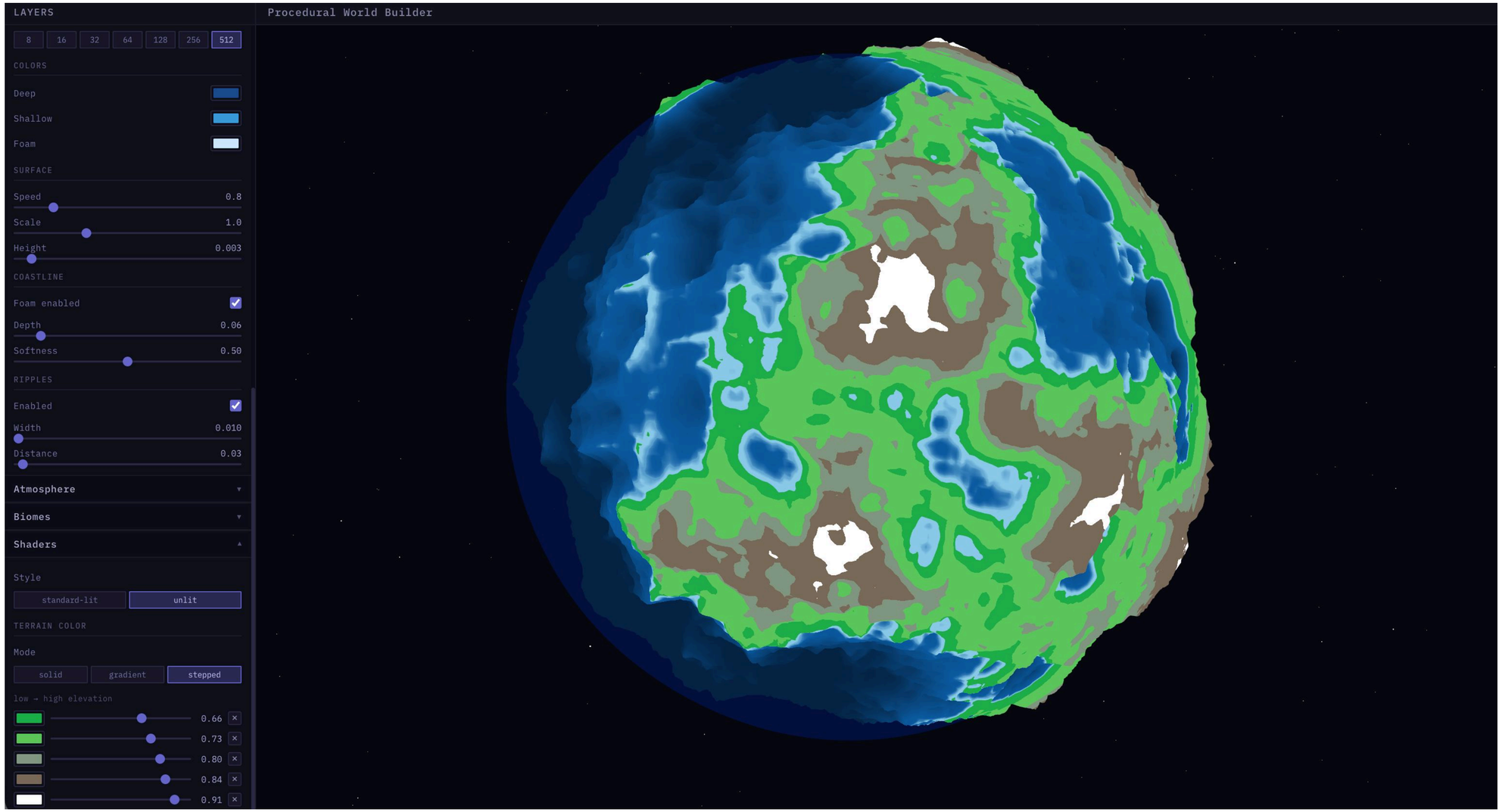
Traditional Web technologies

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React

three.js

Tools:
Web Frameworks
React + Three.js

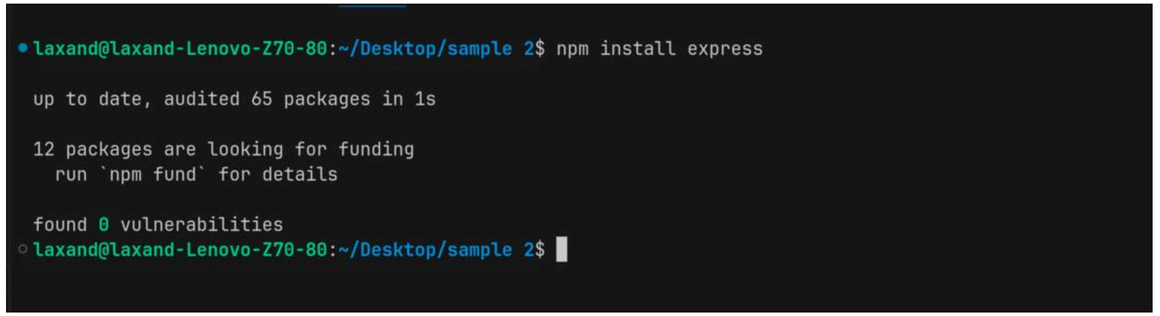


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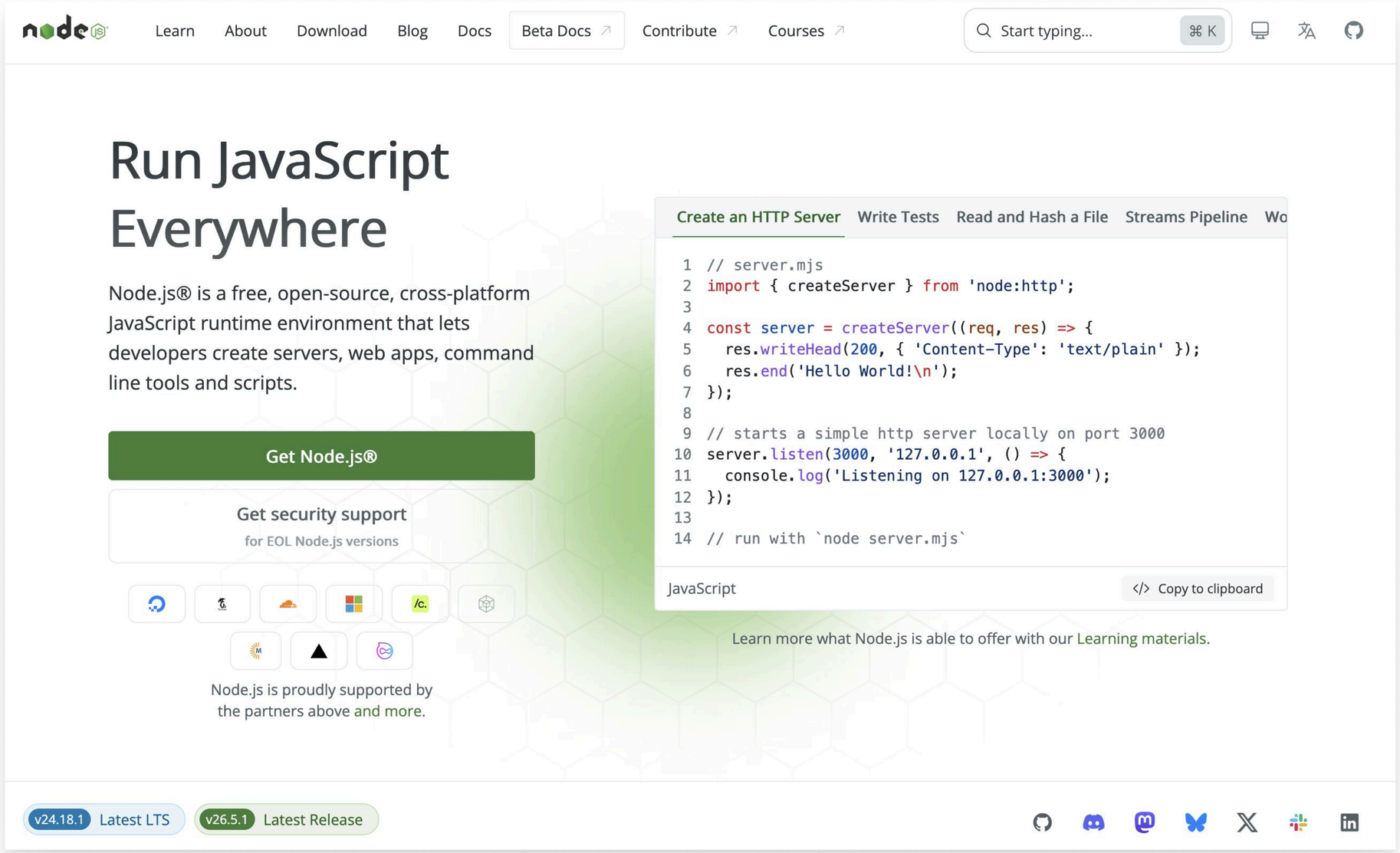


Node.js is a program that lets JavaScript run on your computer (outside a web browser), which is what makes it possible to install and run the tools — like npm and React's build system — that turn your source code into something a browser can load.



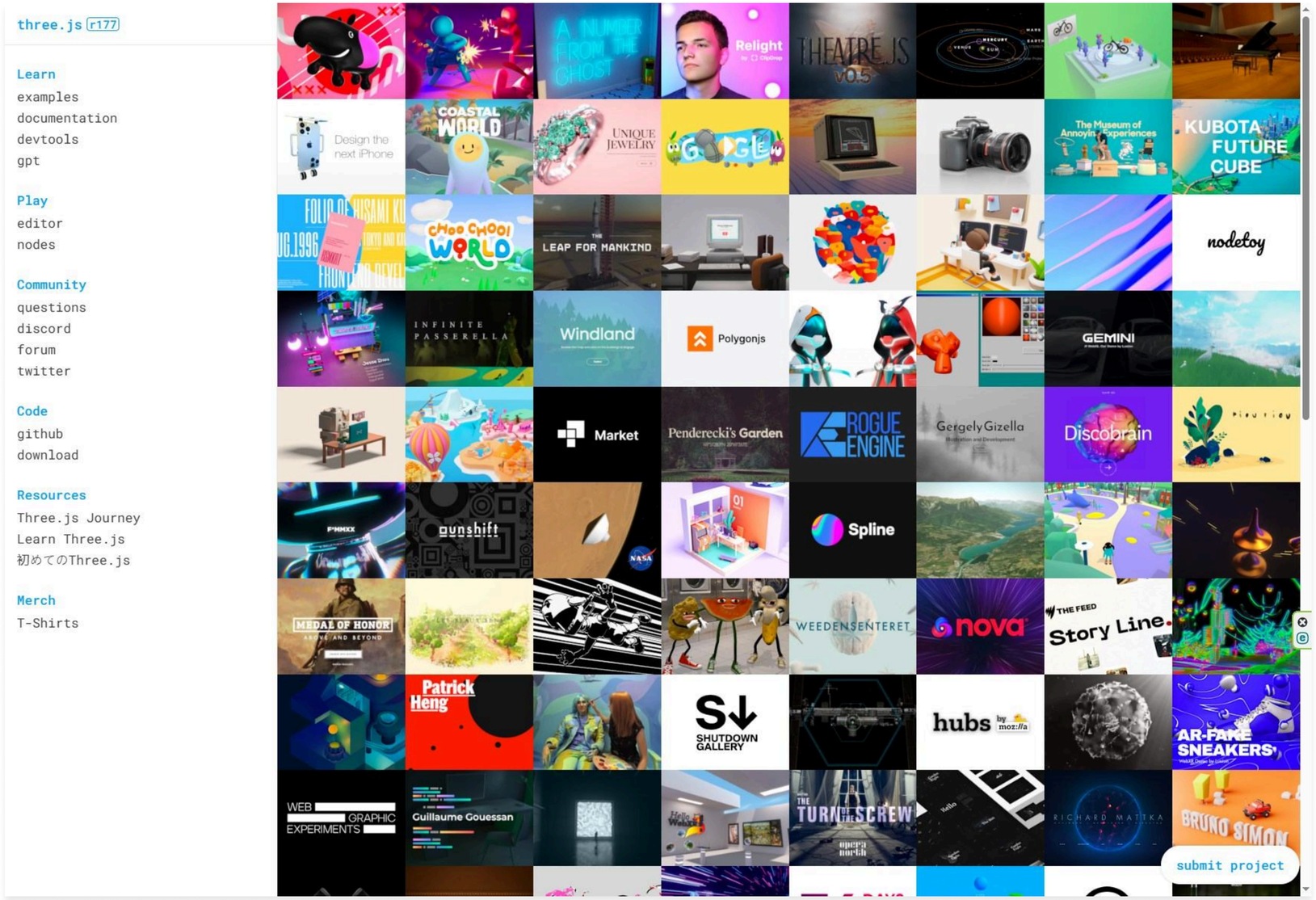
Installing React

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Node.js

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Three.js

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three.js docs manual

en

How to use Post Processing
Matrix Transformations

Basics
Fundamentals
Responsive Design
Prerequisites
Setup

Fundamentals
Primitives
Scenegraph
Materials
Textures
Lights
Cameras
Shadows
Fog
Render Targets
Custom BufferGeometry

Tips
Rendering On Demand
Debugging JavaScript
Debugging GLSL
Taking a screenshot
Prevent the Canvas Being Cleared
Get Keyboard Input From a Canvas
Make the Canvas Transparent
Use three.js as Background in HTML

Optimization
Optimizing Lots of Objects
Optimizing Lots of Objects Animated

Primitives

This article is one in a series of articles about three.js. The first article was about fundamentals. If you haven't read that yet you might want to start there.

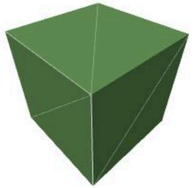
Three.js has a large number of primitives. Primitives are generally 3D shapes that are generated at runtime with a bunch of parameters.

It's common to use primitives for things like a sphere for a globe or a bunch of boxes to draw a 3D graph. It's especially common to use primitives to experiment and get started with 3D. For the majority of 3D apps it's more common to have an artist make 3D models in a 3D modeling program like Blender or Maya or Cinema 4D. Later in this series we'll cover making and loading data from several 3D modeling programs. For now let's go over some of the available primitives.

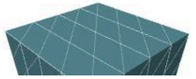
Many of the primitives below have defaults for some or all of their parameters so you can use more or less depending on your needs.

BoxGeometry

A Box



```
const width = 8;
const height = 8;
const depth = 8;
const geometry = new THREE.BoxGeometry( width, height, depth );
```



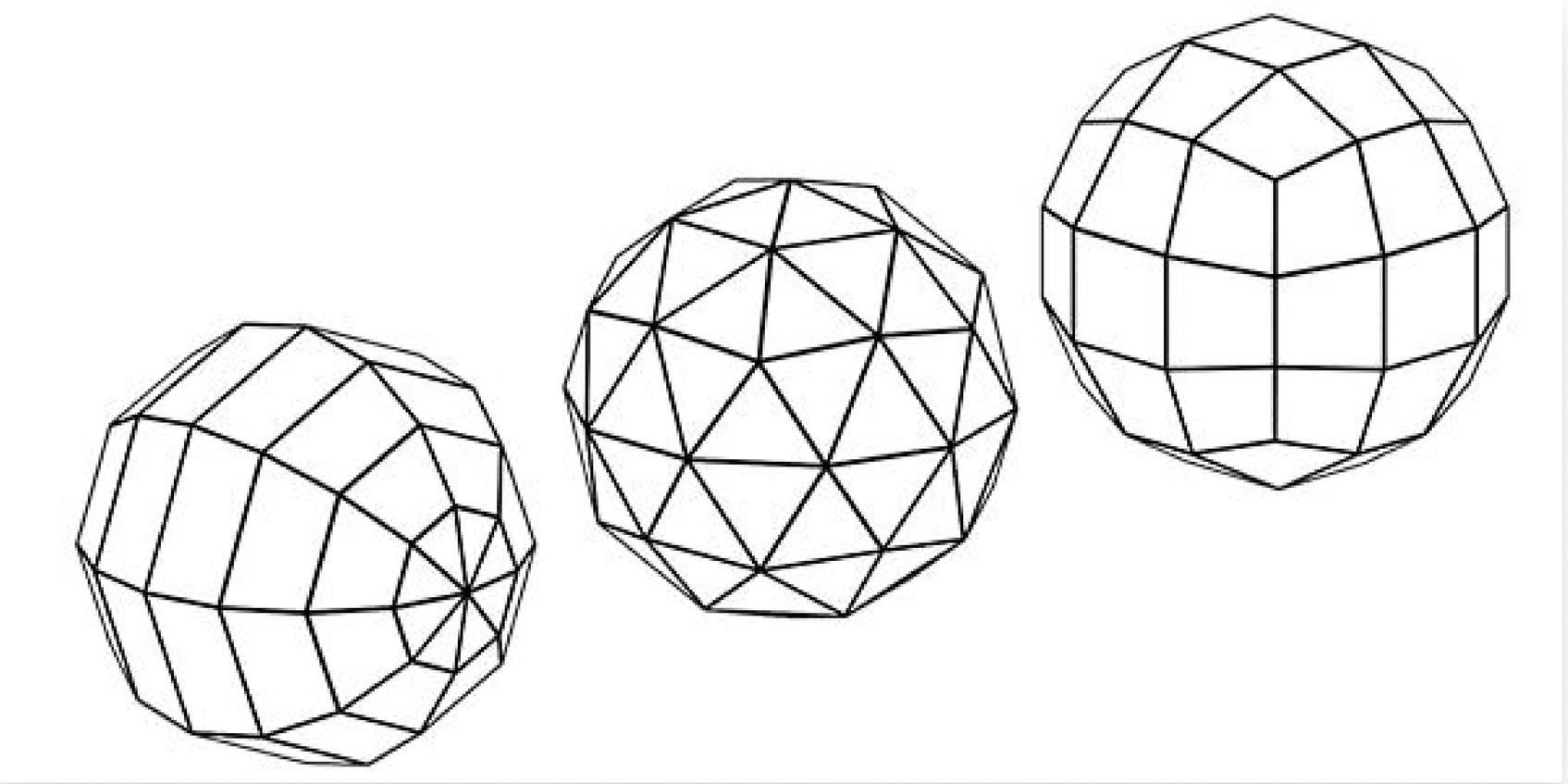
Three.js

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Exercise:
-Create an interactive three.js canvas
-Create some sample geometry in the scene
-Create an orbit camera
-Include a light
-Specify the color and type of material of the objects
-Request to have a “world grid” that fades on distance
-Define the color background of your scene

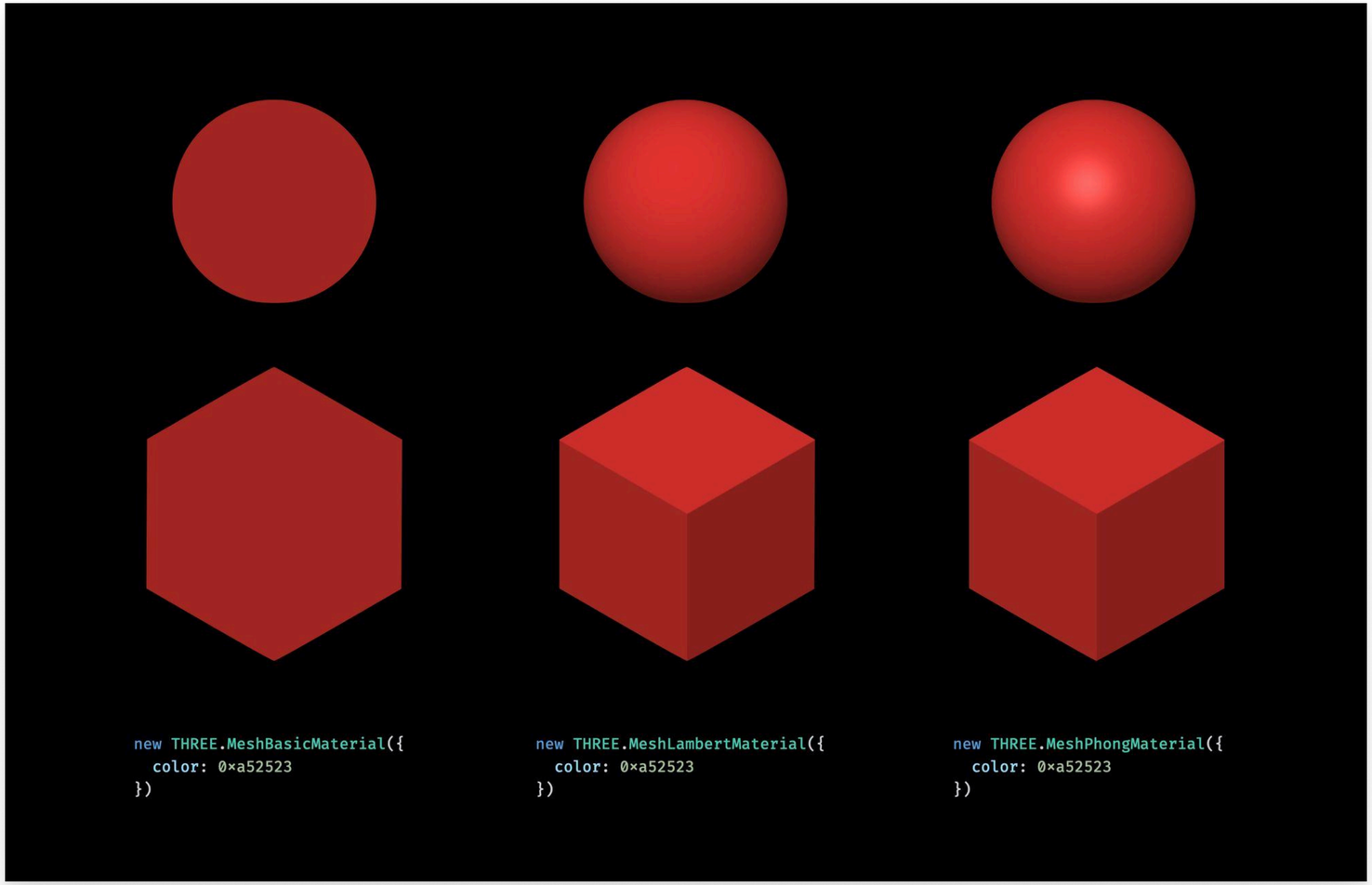
Exercise: Create an interactive 3D canvas using Three.js + React

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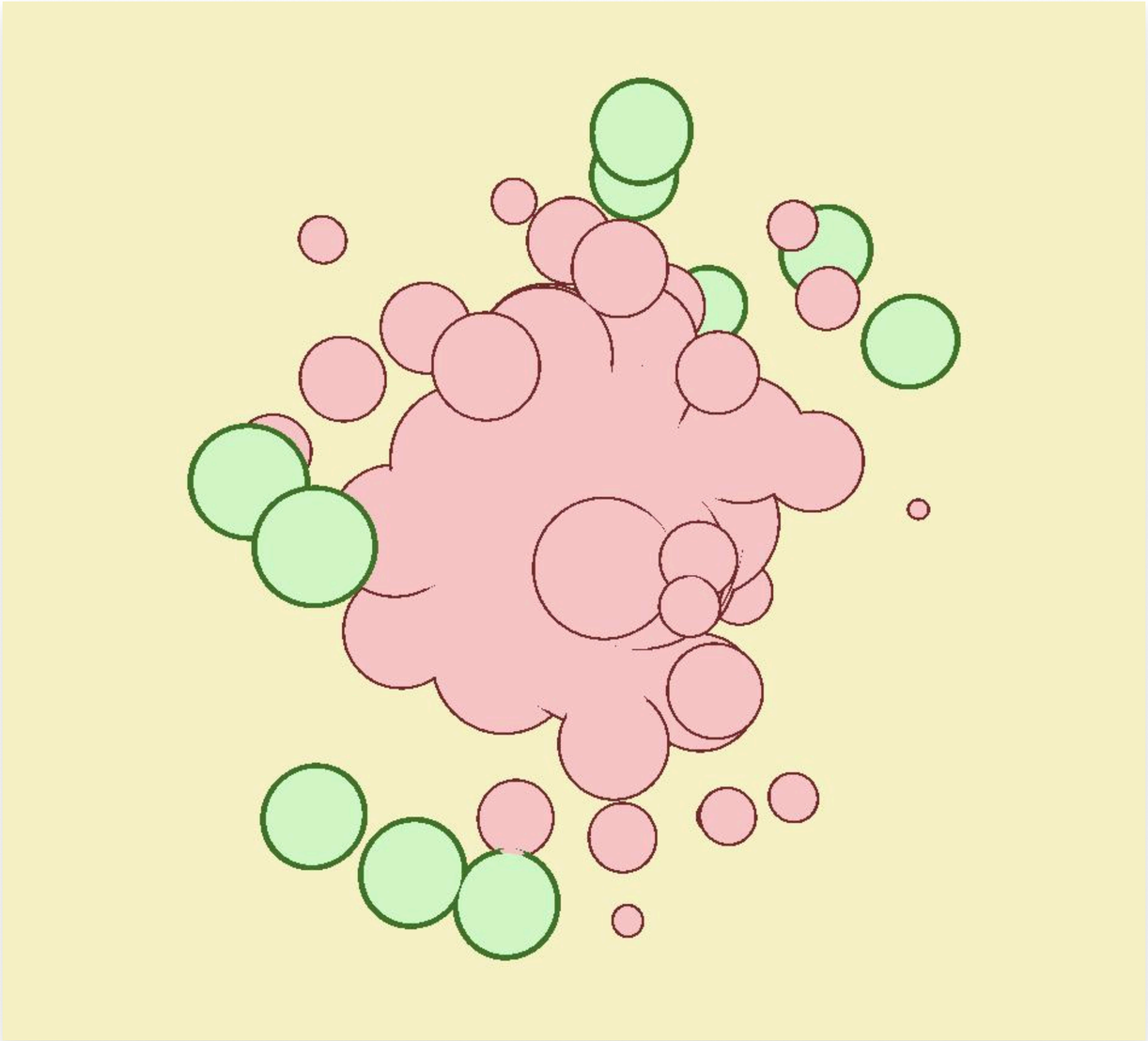
Geometric Construction - topology - ask for “a wireframe visualization as a toggle”

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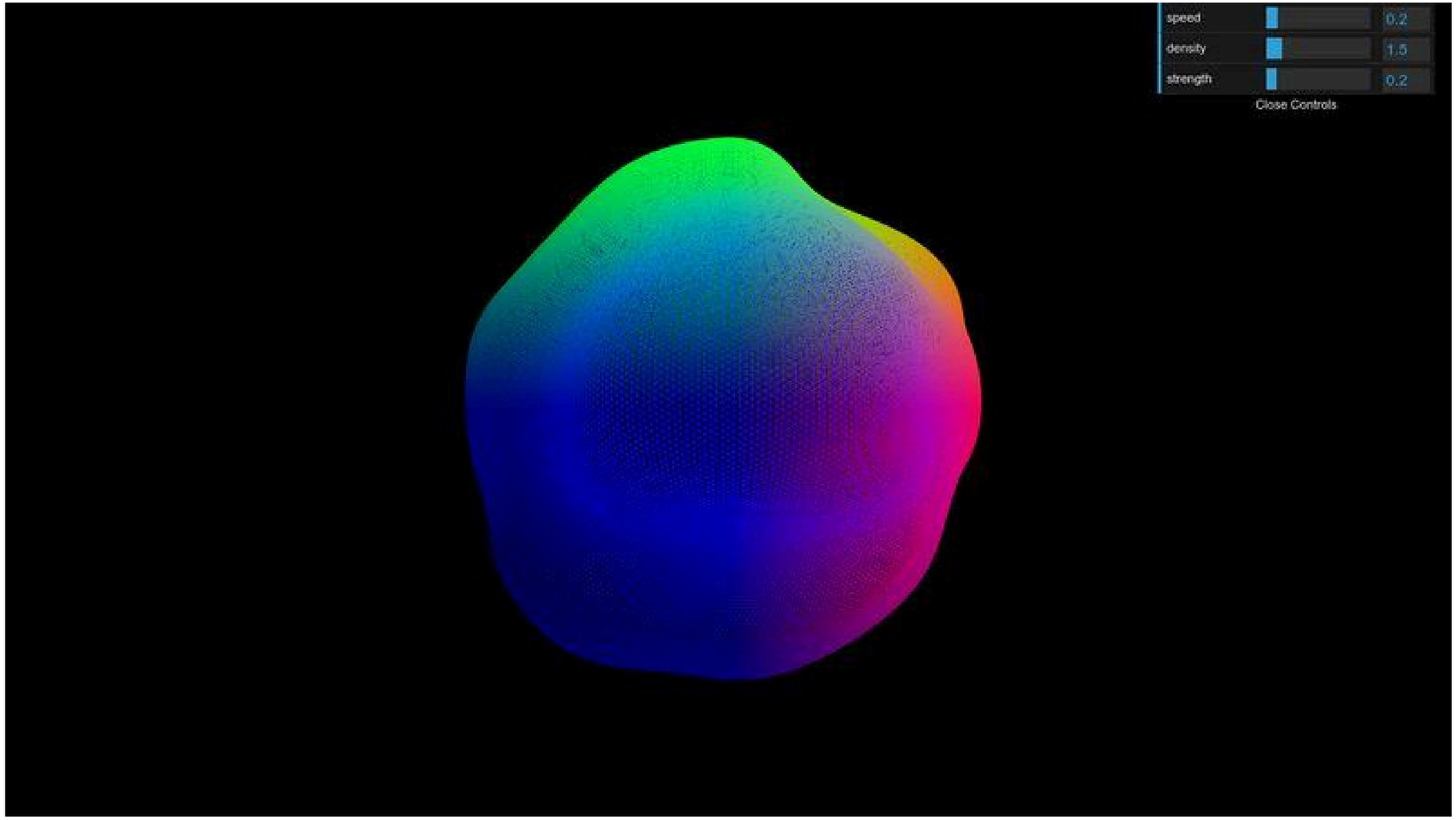
Define the material type

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Shaders

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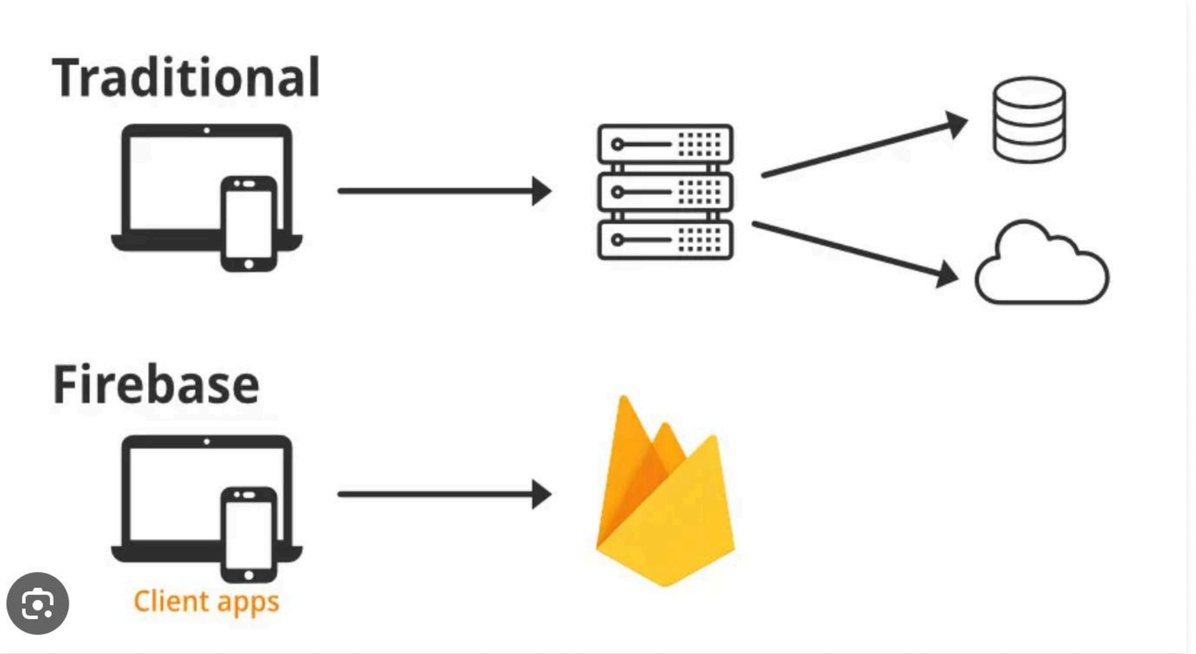
3D Canvas controlled by UI elements

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Backend

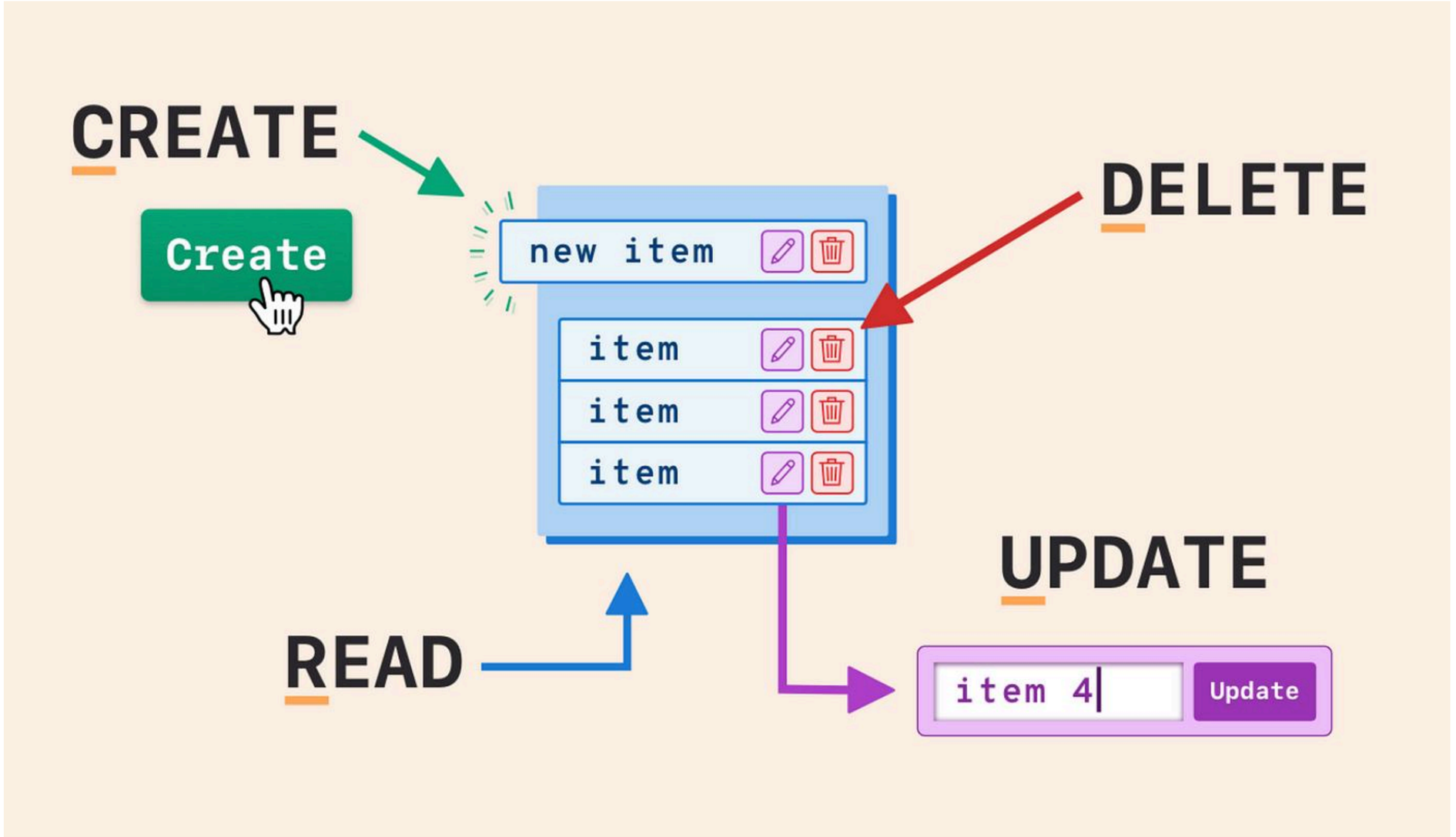
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Back end

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CRUD

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